



Roll Number

JIS UNIVERSITY

2025

END TERM EXAMINATIONS, AY 2024 - 2025(EVEN)

YCS4001 - OPERATING SYSTEMS

Time: 2 Hrs

Maximum Marks : 50

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any Ten (10x1=10 Marks)**

1. In a semaphore, the "wait" operation: (1) CO1 BL1 PO1
 - (a) Decreases the value of the semaphore
 - (b) Increases the value of the semaphore
 - (c) Does not change the semaphore value
 - (d) Terminates the process
2. Virtual memory allows: (1) CO1 BL1 PO1
 - (a) The execution of larger programs than physical memory
 - (b) The use of physical memory only
 - (c) Faster CPU scheduling
 - (d) Better file system organization
3. The Bakery algorithm is used for: (1) CO1 BL1 PO1
 - (a) CPU scheduling
 - (b) Memory management
 - (c) Mutual exclusion in concurrent systems
 - (d) Deadlock prevention
4. Which condition is necessary to prevent deadlocks? (1) CO1 BL1 PO1
 - (a) Hold and wait should be allowed
 - (b) Circular wait should be prevented

- (c) Mutual exclusion should be disabled (d) All resources should be allocated at once
5. The process of swapping is used to:
- (a) Move pages between RAM and disk (b) Improve CPU scheduling
- (c) Improve disk speed (d) Handle deadlocks
6. Real-time operating systems are primarily used for:
- (a) Critical systems with strict deadlines (b) Desktop applications
- (c) Web browsing (d) GAMING
7. Time-sharing systems allow multiple users to:
- a) Share CPU time efficiently
- (a) a) Share CPU time efficiently (b) b) Control hardware directly
- (c) c) Run processes sequentially (d) Avoid using memory
8. Which condition must be satisfied to avoid the critical section problem?
- (a) Mutual Exclusion (b) Progress
- (c) Bounded Waiting (d) All of the above
9. The working set model is used for:
- (a) Deadlock handling (b) Page replacement
- (c) Memory allocation (d) CPU scheduling
10. A page table is used to:
- (a) Manage CPU scheduling (b) Store memory addresses
- (c) Allocate disk space (d) Handle process synchronization
11. Thrashing in an operating system occurs due to:
- (a) Too many processes in the system (b) Too many page faults
- (c) Insufficient CPU scheduling (d) Deadlocks
12. The main advantage of SJF scheduling is:

(1) C01 BL1 PO1

(1) C05 BL1 PO

(1) C02 BL2 PO

(1) C03 BL3 PO

(1) C01 BL1 PO

(1) C02 BL1 PO

(1) C01 BL1 PO

(1) C03 BL1 PO

- (a) a) Minimum average waiting time (b) b) Fastest process execution
(c) c) Less CPU utilization (d) d) Increased turnaround time

Part B**Answer any Two (2x5=10 Marks)**

13. Explain the purpose of Inter-Process Communication (IPC) and describe any two IPC mechanisms. (5) C03 BL6 P01
14. Explain the difference between Contiguous and Non-Contiguous Memory Allocation techniques. (5) C03 BL3 P01
15. Differentiate between Paging and Segmentation in memory management. (5) C04 BL3 P01
16. What is swapping in memory management? Explain its advantages and disadvantages (5) C03 BL3 P01

Part C**Answer any Three (3x10=30 Marks)**

17. Given memory blocks of size 100 MB, 500 MB, 200 MB, 300 MB, 600 MB (in order), allocate the following processes: (10) C04 BL4 P01
1. Process 1: 212 MB
 2. Process 2: 417 MB
 3. Process 3: 112 MB
 4. Process 4: 426 MB

Tasks:

1. Perform memory allocation using First Fit, Best Fit, and Worst Fit techniques.
 2. For each technique, state the final memory block allocation and remaining block sizes.
 3. Identify internal fragmentation if any.
8. Given a page reference string:  (10) C04 BL5 P01
- 5, 3, 4, 2, 1, 4, 3, 5, 1, 2, 4, 5
1. Number of Frames: 3
 2. Tasks:

1. Simulate the FIFO, LRU, optimal page replacement algorithm.
 2. Calculate the number of page faults.
 3. Show the frame content after each page reference.
19. Explain in detail the services, goals, and components of an Operating System (OS).
20. Given:
1. Processes: P0, P1, P2, P3, P4
 2. Resources (A, B, C, D):

(10) CO4 BL3 PC

(10) CO3 BL3 PC

Max instances of Resource type D = 12 (12 allocated + 0 Available)

Given Matrices												
	Allocation Matrix (No of the allocated resources By a process)				Max Matrix Max resources that may be used by a process				Available Matrix Not Allocated Resources			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	0	1	1	0	0	2	1	0	1	5	2	0
P ₁	1	2	3	1	1	6	5	2				
P ₂	1	3	6	5	2	3	6	6				
P ₃	0	6	3	2	0	6	5	2				
P ₄	0	0	1	4	0	6	5	6				
Total	2	12	14	12								

1. Check if the system is in a safe state.
 2. Identify deadlocked processes, if any.
21. Given:
1. Disk Requests: 95, 180, 34, 119, 11, 123, 62, 64
 2. Initial Head Position: 50
 3. Disk Size: 0 to 199
 4. Direction of Head Movement: Towards higher cylinder numbers
 5. Perform disk scheduling using FCFS (First-Come, First-Served), SSTF, C- scan, C-LOOK algorithm.

(10) CO4 BL5

6. Calculate the total head movement.

CO Definitions:

- 1) *To explain the role of an operating system and how it acts as an interface between hardware and software.*
- 2) *To contrast the concepts of processes and threads, and how they are scheduled.*
- 3) *To demonstrate the use of various synchronization tools in solving the critical section problem.*
- 4) *To explain and classify various memory management techniques, including virtual memory.*
- 5) *To apply the knowledge of data structures to explain how file systems can be implemented on secondary storage.*